UNIT - 3

CHARACTERIZATION TECHNIQUES

XRD - SEM - TEM - SPM - AFM - PFM - XPS - XAS - ARPES - DRS - PL - Raman

1. Overview

Characterization reveals the <r>structure, size, shape, composition, surface and physical properties</r> of nanomaterials. Probes used are X-rays, electrons, photons or a sharp physical tip.

Technique	Probe	Information
XRD	X-rays	Crystal structure, phase, lattice parameter, crystallite size, strain
SEM	Electrons (reflect)	Surface morphology, topography, composition (EDX)
TEM	Electrons (transmit)	Internal structure, size, lattice fringes, diffraction (SAED)
STM / AFM / PFM	Sharp tip	3D surface image, atomic resolution, forces, piezo / ferroelectric domains
XPS	X-rays → photoelectrons	Surface composition, oxidation state, chemical bonding
XANES / EXAFS	Tunable X-rays (synchrotron)	Oxidation state, local coordination, bond lengths
ARPES	UV / X-ray → photoelectrons	Band structure E(K), Fermi surface
DRS (UV-Vis)	UV - visible light	Optical absorption, band gap
PL	Light → emission	Defects, recombination, emission energy, quantum yield
Raman	Laser → inelastic scattering	Vibrational modes, phases, defects, graphene layers, CNT

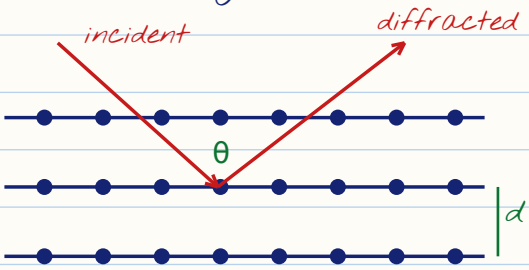
2. X-Ray Diffraction (XRD) and Scherrer Method

X-rays ($\lambda \sim 0.5 - 2.5$ Å, e.g. Cu $K\alpha = 1.5406$ Å) have wavelength comparable to inter-atomic spacing, so a crystal acts as a 3D diffraction grating.

Bragg's law

$$2\, d_{hkl} \sin\theta = n\lambda$$

Constructive interference occurs when the path difference between rays scattered from adjacent planes ($2d \sin\theta$) equals an integer multiple of λ. Here d = interplanar spacing, θ = glancing angle (diffraction angle is 2θ).



Source → sample → detector

Detector moves at 2θ while the sample rotates at θ ($\theta - 2\theta$ geometry)

Output: intensity vs 2θ plot

Fig: Bragg diffraction from crystal planes

Working

- Cu target X-ray tube (40 KV, 30 mA) → Ni filter removes $K\beta$ → monochromatic $K\alpha$ beam falls on the powdered sample on a holder
- Goniometer scans 2θ (typically 10° - 90°, step 0.02°), a scintillation / solid state detector records the intensity
- Each crystalline phase gives a unique set of peaks (fingerprint). Compare with JCPDS / ICDD database cards for phase identification
- Peak position gives d, so lattice parameter. For cubic: $a = d\sqrt{(h^2 + k^2 + l^2)}$
- Peak intensity gives atomic positions / texture; peak width gives crystallite size and strain; amorphous phase gives a broad hump

Scherrer method - crystallite size

Finite crystal size broadens the XRD peaks (in a perfect infinite

crystal peaks are sharp). Scherrer related the mean crystallite size D to the broadening:

$$D = K \lambda / (\beta \cos \theta)$$

- K = shape factor (≈ 0.9 for spherical particles; 0.89 – 1.0), λ = X-ray wavelength, θ = Bragg angle of the peak
- β = FWHM (full width at half maximum) of the peak in **radians** after subtracting the instrumental broadening: $\beta^2 = \beta_{obs}^2 - \beta_{inst}^2$
- Valid for D < ~ 100 nm. Gives the size of coherently diffracting domains, which may be smaller than the particle size

Worked example:

Example: peak at 2θ = 38.2°, FWHM = 0.4° = 0.00698 rad, λ = 1.5406 Å, K = 0.9. D = 0.9 × 1.5406 / (0.00698 × cos 19.1°) = 1.3865 / 0.00659 ≈ 210 Å = 21 nm.

Williamson-Hall analysis (size + strain)

$$\beta \cos \theta = K \lambda / D + 4 \varepsilon \sin \theta$$

Plot β cosθ (y) against 4 sinθ (x): the intercept gives size D, the slope gives micro-strain ε. Other use: dislocation density $\delta = 1/D^2$.

- Advantages of XRD: non-destructive, quick, quantitative phase analysis, needs only mg of sample. Limitation: needs crystalline sample, poor for amorphous, weak for light elements; gives average (bulk) information
- Related: SAXS (small angle) gives particle size 1 – 100 nm; thin film XRD (grazing incidence, GIXRD), rocking curve for epitaxial quality; Rietveld refinement for structure

3. Scanning Electron Microscopy (SEM)

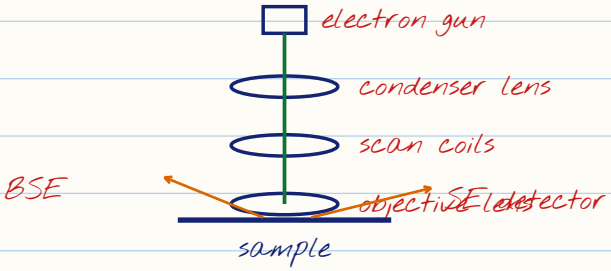
SEM forms an image of the surface by scanning a finely focused electron beam over the sample and collecting the emitted signals. Resolution ~ 1 – 10 nm, magnification 10 – 500,000 ×, large depth of field (3D-like image).

Principle and components

1. **Electron gun** (tungsten filament, LaB_6 or field emission gun FEG)

produces electrons accelerated by 1 - 30 KV.

2. **Condenser lenses** (electromagnetic) demagnify the beam to a spot of 1 - 10 nm; **objective lens** focuses it on the sample.
3. **Scan coils** deflect the beam in a raster pattern over the surface.
4. **Detectors** collect the signals; the display is synchronised with the scan, brightness at each pixel = signal intensity.
5. **High vacuum** (10^{-4} - 10^{-6} Pa) chamber prevents scattering of electrons and contamination.



Signals from the sample:

Secondary electrons (SE, < 50 eV) → topogra
Back-scattered electrons (BSE) → Z contr
Characteristic X-rays → EDX (elemental a
Auger electrons → surface composition
Cathodoluminescence → optical

Fig: Schematic of a scanning electron microscope

- **Secondary electrons** come from the top few nm and give high resolution topographic contrast (edges look bright).
  Back-scattered electrons are high energy electrons reflected by nuclei; the yield increases with atomic number Z so heavy-element regions look bright (composition contrast)
- **EDX / EDS**: the energy of characteristic X-rays identifies elements (qualitative and quantitative, atomic %, mapping)
- **Sample preparation**: dry, vacuum-stable, non-conducting samples are coated with 5 - 10 nm Au / Pt / C by sputtering to avoid charging, mounted on carbon tape / stub. ESEM allows wet samples
- **Uses**: particle shape and size distribution, nanowire length, film thickness (cross-section), grain structure, fracture surfaces, porosity
- Limitation: only surface, 2D picture of a 3D surface, no crystal-structure information, resolution lower than TEM

4. Transmission Electron Microscopy (TEM)

A high energy electron beam (80 - 300 KV) passes through an

ultra-thin sample (< 100 nm), transmitted and diffracted electrons form an image. Resolution < 0.1 nm (atomic), magnification up to 10^6 - 10^7 x.

$$\lambda = h / \sqrt{(2 m e V (1 + eV / 2 m c^2))} \approx 1.226 / \sqrt{V} \text{ nm}$$

At 200 KV, λ = 2.5 pm, far less than the wavelength of light, so atomic resolution is possible.

Working

1. Electron gun (W, LaB_6, FEG) and a stack of condenser lenses produce a parallel beam on the specimen.
2. The specimen (on a 3 mm Cu grid with carbon film) scatters electrons by thickness, density, Z and crystal orientation.
3. The objective lens forms the first image / diffraction pattern; intermediate and projector lenses magnify it onto a fluorescent screen or CCD / CMOS camera.
4. Entire column under high vacuum 10^{-5} - 10^{-7} Pa.

Mode	What is seen	Use
Bright field	Image from direct beam; dense regions dark	Morphology, size distribution
Dark field	Image from a diffracted beam, that crystal bright	Identify particular phases / defects
HRTEM	Phase contrast, lattice fringes	Atomic planes, d-spacing, defects, core-shell
SAED	Diffraction pattern (rings: polycrystal, spots: single crystal)	Crystal structure, orientation
STEM-HAADF	Focused scanning beam, Z-contrast	Atomic columns, EELS / EDX mapping

- Sample preparation: nanoparticles are dispersed in ethanol, sonicated, drop-cast on a carbon-coated grid. Bulk is thinned by mechanical polishing + ion milling, ultramicrotomy or FIB
- Merits: highest resolution, imaging + diffraction + chemistry (EELS) together. Demerits: costly, thin sample needed, electron beam

damage, small area sampled

SEM	TEM
Surface imaging by reflected / emitted electrons	Internal structure by transmitted electrons
1 - 30 KV; resolution ~ 1 - 10 nm	80 - 300 KV; resolution < 0.1 nm
Bulk samples, little preparation	Ultra-thin sample (< 100 nm)
3D-like topography, large depth of field	2D projection image
Cheaper, easier	Expensive, skilled operation

5. Scanning Probe Microscopy (SPM)

A family of microscopes in which a <r>sharp probe (tip, radius 1 - 50 nm) is scanned very close to the surface</r> and a local tip-sample interaction is recorded to build an image. It does not use lenses, so resolution is set by tip sharpness and distance, not by wavelength. It works in air, liquid or vacuum.

Scanning Tunnelling Microscopy (STM) - Binnig and Rohrer 1981

- A metallic tip (W / Pt-Ir) is brought within ~ 0.5 - 1 nm of a conducting surface, a bias voltage (mV - V) causes electrons to **quantum mechanically tunnel** across the vacuum gap
- Tunnelling current: $I \propto V \exp(-2\kappa d)$ with $\kappa = \sqrt{(2m\varphi)}/\hbar$. Current changes by a factor of ~ 10 for each 0.1 nm change in gap (extreme sensitivity), giving vertical resolution of 0.01 nm and lateral of 0.1 nm: individual atoms are imaged
- **Constant current mode**: feedback adjusts the tip height (piezo-scanner) to keep I constant, z-signal = topography. **Constant height mode**: fast, records I
- Limitation: sample must be conducting / semiconducting; STS (spectroscopy) measures the local density of states

Common features of SPM

- Piezoelectric scanner (tube) gives sub-Å movement in x, y, z
- Feedback loop (PID) controls the tip-surface interaction

- Vibration isolation is essential, image formed pixel by pixel line by line
- Variants: AFM (force), MFM (magnetic force), EFM (electrostatic), KPFM (surface potential), SNOM / NSOM (near-field optical), SThM (thermal), PFM (piezo response), SECM, conductive AFM

6. Atomic Force Microscopy (AFM)

Invented by Binnig, Quate and Gerber (1986). Measures the <r>force between a sharp tip and the surface</r>, works on insulators as well, so it is more versatile than STM.

Construction and working

- **Cantilever** (Si or Si_3N_4, 100 – 200 μm long, spring constant 0.1 – 50 N/m) with an integrated sharp tip (radius 5 – 20 nm)
- **Laser and photodiode detector**: a laser beam reflects off the back of the cantilever onto a four-quadrant photodiode; tiny deflection is magnified into a large spot movement (optical lever)
- **Piezo scanner** moves the sample (or tip) in the x-y raster; **feedback loop** keeps the deflection or oscillation amplitude constant and the z voltage is recorded as the height image
- Force obeys Hooke's law: $F = -k\,z$. Forces: van der Waals (attractive at long range), Pauli repulsion (short range), capillary, electrostatic, magnetic

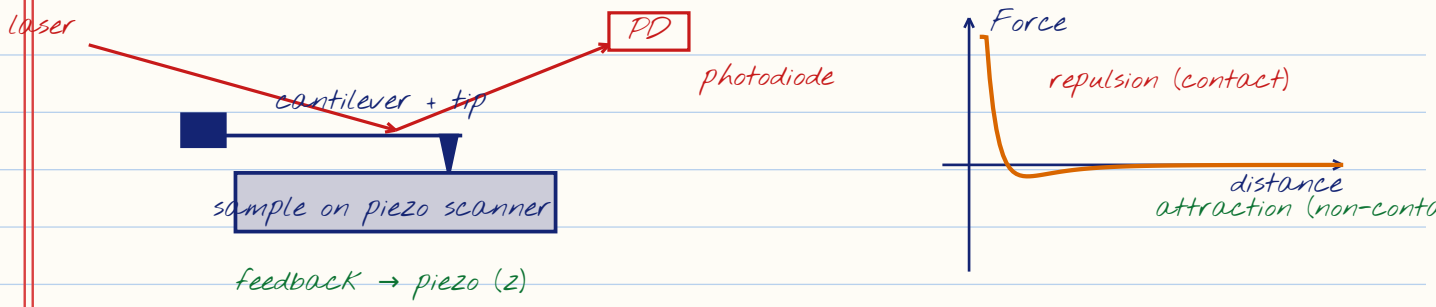


Fig: AFM principle (optical lever) and force-distance curve

Mode	Tip-sample	Features
Contact	Tip touches surface, repulsive force	Fast, high resolution; may damage soft samples, lateral forces

Non-contact	Tip 1 – 10 nm above, attractive vdW force, cantilever oscillates near resonance	No wear; lower resolution, can be affected by water layer
Tapping (intermittent)	Oscillating cantilever lightly taps the surface, amplitude is the feedback signal; phase image shows material contrast	Most popular; minimal damage; suits polymers, DNA, soft nanoparticles

- Information: 3D topography, roughness (Ra, RMS), particle / grain size and height, step height of films (sub-nm), mechanical (nano-indentation, modulus), adhesion, friction (LFM), conductivity (c-AFM), magnetic domains (MFM), surface potential (KPFM)
- **Advantages**: no vacuum or conducting sample needed, can work in liquid (biological samples), true 3D height data. **Disadvantages**: small scan area (~ 100 μm), slow, tip artefacts (convolution), image drift

STM	AFM
Measures tunnelling current	Measures force / cantilever deflection
Conducting samples only	Conducting and insulating samples
Needs vacuum (usually)	Air, liquid or vacuum
Atomic resolution routine	Atomic resolution in non-contact UHV; typically nm

7. Piezoresponse Force Microscopy (PFM)

PFM is an AFM-based (contact mode) technique used to image and manipulate ferroelectric and piezoelectric domains at the nanoscale. It uses the converse piezoelectric effect: a material deforms when an electric field is applied.

- A conductive tip (Pt / Ir-coated) in contact with the sample acts as the top electrode. An AC voltage $V = V_0 \cos \omega t$ (+ DC) is

applied between tip and the bottom electrode

- The sample expands / contracts: $\Delta z = d_{33} V_0 \cos(\omega t + \varphi)$. The cantilever follows this tiny vibration (pm range) and a lock-in amplifier extracts the amplitude and phase
- **Amplitude** → magnitude of piezoelectric coefficient d_{33} (domain strength; zero at domain walls). **Phase** → polarisation direction: in-phase (0°) for P pointing down, out-of-phase (180°) for P pointing up
- **Vertical PFM** detects out-of-plane polarisation; **lateral PFM** (torsion of the cantilever) detects in-plane polarisation
- **Local polarisation switching**: sweeping a DC bias gives piezoresponse hysteresis loops (coercive voltage, imprint), one can **write domains** by applying a bias above the coercive voltage
- **Applications**: ferroelectric thin films (PZT, $BiFeO_3$, $BaTiO_3$), nanostructured ferroelectrics for FeRAM, multiferroics, piezo nanowires (ZnO), biological piezoelectricity (collagen)
- **Limitation**: electrostatic artefacts, tip wear, needs conductive tip; Dual-AC resonance tracking (DART) improves signal

8. X-ray Photoelectron Spectroscopy (XPS / ESCA)

XPS is a surface sensitive technique (top 1 - 10 nm) that identifies the elements, their chemical state and composition from the photoelectrons emitted when the sample is irradiated by soft X-rays (Al Kα 1486.6 eV or Mg Kα 1253.6 eV). It is based on the photoelectric effect.

$$E_B = h\nu - E_K - \varphi_{spec}$$

- E_B = binding energy of the core level; E_K = measured kinetic energy; φ = spectrometer work function. E_B is a signature of each element and orbital (1s, $2p_{3/2}$ ...)
- **Setup**: X-ray source + monochromator → sample in UHV (10^{-9} torr) → electron energy analyser (hemispherical) → detector. UHV needed because electrons are scattered by gas molecules and the surface would be contaminated
- **Survey scan** (0 - 1100 eV) identifies all elements except H and He;

high-resolution scan of a peak gives chemical state

- **Chemical shift**: change in E_B (0.1 – 10 eV) with oxidation state / bonding; e.g. C 1s: C–C 284.8 eV, C–O 286.5, C=O 288, O–C=O 289 eV; Ti $2p_{3/2}$ 458.8 eV is Ti^{4+}; Fe^{2+} / Fe^{3+} distinguished
- **Spin-orbit splitting**: p, d, f levels split into doublets ($2p_{3/2}/2p_{1/2}$ area ratio 2:1, d 3:2, f 4:3)
- **Quantification**: atomic % ∝ peak area / sensitivity factor. Depth profile by Ar^+ ion sputtering; **angle-resolved XPS** for thin film thickness; XPS mapping; also valence band edge
- Charge referencing with adventitious C 1s = 284.8 eV (insulating samples); other features: shake-up satellites, Auger lines
- Uses in nano: surface oxidation, doping, core-shell composition, functionalisation of CNT / graphene (C–O, C–N), catalysts, thin film interfaces
- Limitations: surface only, UHV, cannot detect H / He, expensive, X-ray damage to some samples

9. XANES and XAFS (X-ray Absorption Spectroscopy)

When X-ray energy is scanned across the absorption edge of an element, the absorption coefficient $\mu(E)$ rises sharply (edge) as a core electron is ejected, followed by oscillations. Needs a **synchrotron** source with a tunable, intense X-ray beam. It probes the <r>local atomic and electronic structure</r> of a chosen element, even in amorphous materials, solutions or very dilute nano systems.

Region	Energy range	Information
Pre-edge	Few eV before edge	Forbidden 1s → 3d transitions; coordination symmetry (tetrahedral vs octahedral)
XANES (NEXAFS)	Edge to ~ 50 eV above	Oxidation state (edge shifts to higher E with higher valence), coordination geometry, unoccupied DOS, covalency

EXAFS	~ 50 – 1000 eV above edge	Bond distance, coordination number, disorder (Debye-Waller factor) of neighbouring shells

XAFS = XANES + EXAFS. In EXAFS the outgoing photoelectron wave scatters off neighbouring atoms and interferes with itself, modulating $\mu(E)$:

$$\chi(k) = \sum_j N_j \, S_0^2 \, f_j(k) \,/\, (k \, R_j^2) \cdot \sin(2kR_j + \delta_j) \cdot e^{-2\sigma_j^2 k^2} \, e^{-2R_j/\lambda}$$

- Fourier transform of $k^3\chi(k)$ gives a radial distribution function with peaks at the neighbour distances R_j
- Modes: transmission (concentrated samples), fluorescence yield (dilute / thin film), electron yield (surface). Edges: K, L_{III}
- Uses: oxidation state and site of dopants in nano oxides (Mn in ZnO), size-dependent bond contraction, active sites in nanocatalysts, in-situ battery studies
- Advantage: element specific, no long-range order needed. Limitation: needs synchrotron, data analysis is complex (Athena / Artemis)

10. Angle Resolved Photoemission Spectroscopy (ARPES)

ARPES is the direct experimental probe of the <r>electronic band structure $E(k)$<l/r>: by measuring the kinetic energy and emission angle of photoelectrons ejected by UV / soft X-ray photons (He I 21.2 eV, synchrotron 20 – 100 eV, lasers 6 eV).

- Energy conservation: $E_B = h\nu - \varphi - E_K$. Momentum parallel to the surface is conserved when the electron leaves the crystal:

$$\hbar \, k_{\parallel} = \sqrt{(2 \, m \, E_K)} \sin\theta$$

- A hemispherical analyser with a 2D detector records intensity vs E_K and angle θ in one shot, so E_B vs $k_{\parallel}$ (band dispersion) is mapped. Rotating the sample gives the full Fermi surface
- Needs UHV ($< 10^{-10}$ torr), clean, flat, well-ordered single-crystalline surface prepared by in-situ cleaving / sputtering-annealing; low temperature for sharp spectra
- Photon energy variation gives k_z dependence (3D bands),

spin-resolved ARPES shows spin texture

- Uses: band structure and effective mass, Fermi surface, superconducting gap (cuprates, iron pnictides), Dirac cone of graphene and topological insulators (Bi_2Se_3), quantum well states in thin films, band bending
- Limitation: very surface sensitive (~ 0.5 – 1 nm), UHV and single crystals, matrix elements and photon energy dependence

II. Diffuse Reflectance Spectra (DRS)

UV-Vis DRS measures light diffusely scattered by a **powder / opaque** sample (transmission is impossible) using an integrating sphere coated with $BaSO_4$ / Spectralon. The reflectance R (λ) vs wavelength is converted into an absorption-like function and used to find the optical band gap.

Kubelka-Munk function

$$F(R) = (1 - R)^2 / 2R = K / S$$

K = absorption coefficient, S = scattering coefficient, R = diffuse reflectance of an infinitely thick sample relative to a white standard.

Tauc plot for the band gap

$$(F(R)\,h\nu)^{1/n} = A\,(h\nu - E_g)$$

- n = 1/2 for direct allowed transition (plot $(F(R)h\nu)^2$ vs $h\nu$); n = 2 for indirect allowed transition (plot $(F(R)h\nu)^{1/2}$ vs $h\nu$). $h\nu = 1240/\lambda(nm)$ eV
- Extrapolate the linear region of the plot to the $h\nu$ axis: intercept = E_g
- Example: TiO_2 anatase $E_g \approx 3.2$ eV (absorption edge ~ 387 nm), ZnO ≈ 3.3 eV; CdS ≈ 2.4 eV
- Blue shift of the absorption edge in smaller particles confirms quantum confinement, red shift / tail absorption shows doping, oxygen vacancies (Urbach energy)
- Uses: photocatalysts, pigments, solar absorbers, dye-sensitised

materials. Limitations: scattering effects, surface reflection, assumptions in K-M theory

12. Photoluminescence (PL) Spectra

PL is the light emitted by a material after it has absorbed photons: **excitation → relaxation → recombination → emission**. It is a contactless and non-destructive probe of the electronic structure, defects and impurities.

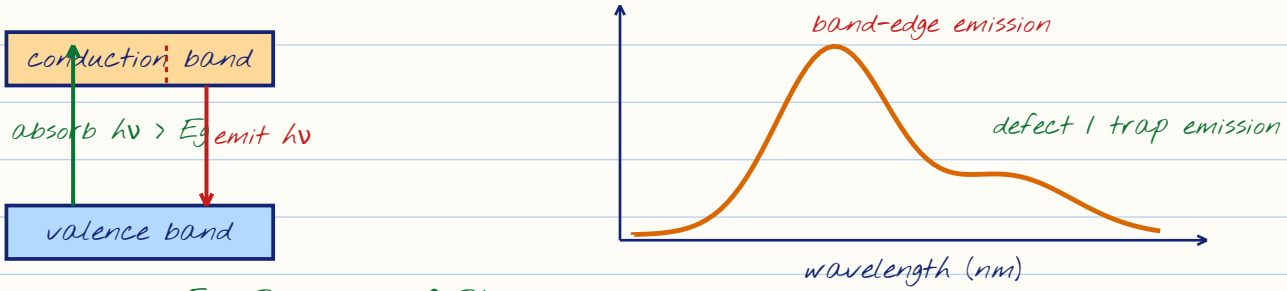


Fig: Principle of PL and a typical emission spectrum

- **Instrument**: Xe lamp / laser (e.g. 325 nm He-Cd) → excitation monochromator → sample → emission monochromator → PMT / CCD detector. Excitation spectrum (fixed emission) and emission spectrum (fixed excitation) are recorded
- **Band-edge (near band edge, NBE) emission** gives E_g; its blue shift shows quantum confinement, FWHM gives size distribution. Stokes shift = absorption − emission energy
- **Defect emission**: oxygen vacancies / zinc interstitials in ZnO give a green-yellow band at 500 – 600 nm; the ratio $I_{UV}/I_{visible}$ measures crystal quality
- **Quenching** of PL (e.g. in TiO_2 / graphene composites) indicates efficient charge separation useful for photocatalysis
- **Time-resolved PL (TRPL)** gives carrier lifetime τ; **quantum yield** = emitted / absorbed photons (> 90 % for CdSe / ZnS dots); temperature-dependent PL gives activation energies and exciton binding energy
- Uses: quantum dots, LED phosphors, solar cell materials (perovskites), biological labels, sensors, defect studies
- Limitation: non-radiative processes are invisible; mostly qualitative

for impurities, interpretation needs care

13. Raman Spectroscopy

Raman spectroscopy is based on <r>inelastic scattering of monochromatic light (laser)</r> by molecular / lattice vibrations (C. V. Raman, 1928, Nobel Prize 1930). The scattered photon gains or loses the energy of a vibration (phonon).

Process	Photon energy change	Intensity
Rayleigh	Elastic, no change	Strongest (blocked by a notch / edge filter)
Stokes Raman	Photon loses energy $\hbar\Omega$ (molecule excited)	Weak ($\sim 10^{-6}$), strong at room temperature
Anti-Stokes Raman	Photon gains energy (from excited vibrational state)	Weaker; rises with temperature, ratio gives T

$$\text{Raman shift } (cm^{-1}) = 1/\lambda_0 - 1/\lambda_s$$

- Selection rule: the vibration must change the **polarisability** of the bond / crystal (Raman active). It complements IR spectroscopy (needs a change in dipole moment); for centro-symmetric molecules mutual exclusion applies
- **Instrument**: laser (488, 514, 532, 633, 785 nm) → microscope → sample → notch filter → spectrometer (grating) → CCD. Micro-Raman spot ~ 1 μm; no sample preparation; works through glass / in water
- Resonance Raman and SERS (surface enhanced, on Au / Ag nanostructures, enhancement 10^6 – 10^{14}, up to single-molecule detection) and TERS give very high sensitivity

Use in nanomaterials

- **Carbon nanotubes**: RBM (radial breathing mode, 100 – 350 cm^{-1}) gives diameter $d(nm) \approx 248/\omega_{RBM}$; G band ~ 1580 cm^{-1} (sp^2); D band ~ 1350 cm^{-1} (defects, I_D/I_G ratio); 2D band ~ 2700 cm^{-1}
- **Graphene**: layer number from the shape of the 2D band (single sharp Lorentzian for monolayer; $I_{2D}/I_G \approx 2$); doping and strain

from G-peak shifts

- **Phonon confinement**: in Si, TiO_2, CdSe nanocrystals the Raman peak shifts and broadens asymmetrically as size decreases; identifies phases (anatase E_g 144 cm^{-1}, rutile 447 / 612 cm^{-1}), stress and defects
- **Others**: MoS_2 layer thickness (E_{2g} – A_{1g} separation), polymorphs of carbon (diamond sharp line at 1332 cm^{-1}), strain in films, composition of alloys
- **Limitations**: fluorescence background, weak signal, laser heating / damage of small particles

Quick Revision – Important Questions

1. Derive Bragg's law. Explain how XRD gives crystallite size by the Scherrer method (with example).
2. Explain the construction and working of SEM; distinguish SEM and TEM.
3. Describe TEM with modes (bright field, HRTEM, SAED) and sample preparation.
4. What is scanning probe microscopy? Explain STM and AFM with diagrams and modes.
5. Explain piezoresponse force microscopy.
6. Write short notes on XPS, XANES / EXAFS and ARPES.
7. Explain DRS and the Tauc plot method for finding the band gap.
8. Explain photoluminescence and Raman spectroscopy with uses in nano-materials.